THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 5 & 6: **Circles, Measuring Space: Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 16 cm	half-chord $= \sqrt{10^2 - 6^2} = 8$, so chord $= 2 \times 8 = 16$
2	**(b)** 10 cm	half-chord $= \sqrt{13^2 - 12^2} = 5$, so chord $= 2 \times 5 = 10$
3	**(b)** 17 cm	$r = \sqrt{15^2 + 8^2} = 17$
4	**(d)** 5 cm	$d = \sqrt{13^2 - 12^2} = 5$
5	**(c)** 35°	angle at the centre is double the angle on the circle
6	**(b)** 132 cm	$2\pi r = 2 \times \frac{22}{7} \times 21 = 132$
7	**(c)** 14 cm	$r = \frac{C}{2\pi} = \frac{88 \times 7}{44} = 14$
8	**(c)** 33 cm	$\frac{90}{360} \times 2\pi r = 33$
9	**(b)** 22 cm	$\frac{30}{360} \times 2\pi r = 22$
10	**(d)** 60 cm^2	$s = 20$, area $= \sqrt{20 \cdot 12 \cdot 5 \cdot 3} = 60$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$OA = OB$ and $\angle AOB = 60^\circ \Rightarrow \triangle OAB$ is equilateral, so $AB = 12$ cm.	1+1
12	$d = \sqrt{13^2 - 5^2} = 12$ cm for each; equal chords are equidistant from the centre.	1+1
13	$s = 6$; area $= \sqrt{6 \times 3 \times 2 \times 1} = 6$ cm^2. Right-angled: $\frac{1}{2} \times 3 \times 4 = 6$ ✓	1+1
14	$bh = 96 \Rightarrow h = \frac{96}{12} = 8$ cm.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$6x + 12 = 180 \Rightarrow x = 28$ (2). $\angle A = 108^\circ, \angle C = 72^\circ$ (1).	2+1
16	Radii are in the ratio $5 : 4$ (1). Areas πr^2 in the ratio $25 : 16$ (2).	1+2
17	In a parallelogram opposite angles are equal: $\angle A = \angle C$ (1). In a cyclic quadrilateral $\angle A + \angle C = 180^\circ$ (1). So each is $90^\circ \Rightarrow$ rectangle (1).	1+1+1
	OR: $\angle A + \angle D = 180^\circ$ (co-interior) and $\angle A + \angle C = 180^\circ$ (cyclic) $\Rightarrow \angle C = \angle D$ (2). Base angles equal $\Rightarrow AD = BC$ (1).	2+1
18	$r = 14$ m (1); area $= \frac{22}{7} \times 196 = 616$ m^2 (1); cost $= 88 \times 12 = ₹1056$ (1).	1+1+1
	OR: $\pi(14^2 - 7^2) = \frac{22}{7} \times 147 = 462$ cm^2.	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 80° (1). (ii) $OB = OC \Rightarrow \angle OBC = 50^\circ$ (1). (iii) Same segment $\Rightarrow \angle BDC = 40^\circ$ (1). (iv) Half-chord $= \sqrt{60^2 - 36^2} = 48$; $BC = 96$ cm (2).	1+1+1+2
	OR: Let $OM = ON$ with $OM \perp AB, ON \perp CD$. Right triangles OMA, ONC: $OA = OC$, $OM = ON \Rightarrow$ congruent (RHS) $\Rightarrow AM = CN$ (2). Halves equal $\Rightarrow AB = CD$ (1). Length $= 2\sqrt{169 - 144} = 10$ cm (2).	2+1+2
20	Inscribed hexagon: six equilateral triangles of side $r \Rightarrow$ perimeter $6r < 2\pi r \Rightarrow \pi > 3$ (2). Circumscribed hexagon: side $= \frac{2r}{\sqrt{3}} \Rightarrow$ perimeter $4\sqrt{3}r > 2\pi r \Rightarrow \pi < 2\sqrt{3} \approx 3.46$ (2). More sides squeeze the two perimeters closer, trapping π more tightly (1).	2+2+1
	OR: (i) $AB = AC = BC = r$ (radii) (1). (ii) $\angle CAB = \angle BAD = 60^\circ$, so $\angle CAD = 120^\circ$ (2). (iii) Each outer arc subtends 240° at its centre: total $= 2 \times \frac{240}{360} \times 2\pi r = \frac{8\pi r}{3}$ (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 5 & 6: **Circles, Measuring Space: Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 75°	angle at the centre is double the angle on the circle
2	**(a)** 115°	opposite angles of a cyclic quadrilateral add to 180°
3	**(d)** 90°	angle in a semicircle
4	**(c)** exactly one	unique circumcircle
5	**(d)** 180°	Theorem 11
6	**(d)** 120 cm^2	area $= bh$
7	**(b)** 60 cm^2	area $= \frac{1}{2}bh$
8	**(d)** 154 cm^2	$\pi r^2 = \frac{22}{7} \times 7^2 = 154$
9	**(c)** the same for all circles (π)	the C/D ratio is the constant π
10	**(d)** the semi-perimeter of the triangle	$s = \frac{a+b+c}{2}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Chord at 3 cm: $2\sqrt{25-9} = 8$ cm; chord at 4 cm: $2\sqrt{25-16} = 6$ cm. The chord nearer the centre is longer.	1+1
12	The four points are concyclic (Theorem 10): they lie on one circle.	2
13	A median divides a triangle into two triangles of equal area (equal bases, same height): $\text{ar}(ABD) = 24$ cm^2.	1+1
14	The square with side 10 units; area 100 sq. units (e.g. $9 \times 11 = 99 < 100$).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\angle BDC = \angle BAC = 30^\circ$ (angles in the same segment) (1+1). $\angle CBD = 180^\circ - 100^\circ - 30^\circ = 50^\circ$ (1).	2+1
16	Circumference $= 176$ cm (1). 1.76 km $= 176000$ cm (1). Revolutions $= \frac{176000}{176} = 1000$ (1).	1+1+1
17	Half-chord $= \sqrt{r^2 - d^2}$ (Baudhāyana–Pythagoras) (1); chord $= 2\sqrt{r^2 - d^2}$ (1). For $r = 10, d = 6$: $2\sqrt{64} = 16$ (1).	1+1+1
	OR: At 6 cm: $2\sqrt{100-36} = 16$ cm; at 8 cm: $2\sqrt{100-64} = 12$ cm (2). $16 > 12$ and $6 < 8$: the chord nearer the centre is longer (1).	2+1
18	Area $= 20 \times 9 = 180$ cm^2 (1+1). Using the 15 cm side as base: $15 \times h = 180 \Rightarrow h = 12$ cm (1).	2+1
	OR: Area $= \frac{1}{2}(12 + 20) \times 7 = 112$ cm^2.	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$CA = CD \Rightarrow \angle ACE = 2\angle ADC$ (exterior angle) (2). $CB = CD \Rightarrow \angle BCE = 2\angle BDC$ (2). Subtract: $\angle ACB = \angle ACE - \angle BCE = 2(\angle ADC - \angle BDC) = 2\angle ADB$ (1).	2+2+1
	OR: (i) Both are angles in a semicircle: 90° each (2). (ii) $180 - 90 - 30 = 60^\circ$ (1). (iii) $180 - 90 - 50 = 40^\circ$ (1). (iv) ACBD has all four vertices on the circle, so it is cyclic (1).	2+1+1+1
20	(i) $\frac{360}{8} = 45^\circ$ (1). (ii) $\frac{45}{360} \times 2\pi \times 14 = 11$ cm (1). (iii) $11 + 14 + 14 = 39$ cm (1). (iv) $\frac{45}{360} \times \frac{22}{7} \times 196 = 77$ cm^2 (2).	1+1+1+2
	OR: (i) $s = 88$ m (1). (ii) $\sqrt{63 \times 49 \times 36 \times 28} = 1764$ m^2 (2). (iii) $25^2 + 60^2 = 65^2$ and $39^2 + 52^2 = 65^2$, so both triangles are right-angled; areas $750 + 1014 = 1764$ ✓ (1). (iv) $1764 \times 20 = ₹35{,}280$ (1).	1+2+1+1

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ANSWER KEY & MARKING SCHEME SET C

Chapters 5 & 6: **Circles, Measuring Space: Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 30 cm	half-chord $= \sqrt{17^2 - 8^2} = 15$, so chord $= 2 \times 15 = 30$
2	**(b)** 12 cm	half-chord $= \sqrt{10^2 - 8^2} = 6$, so chord $= 2 \times 6 = 12$
3	**(a)** 25 cm	$r = \sqrt{24^2 + 7^2} = 25$
4	**(c)** 15 cm	$d = \sqrt{17^2 - 8^2} = 15$
5	**(b)** 50°	angle at the centre is double the angle on the circle
6	**(b)** 22 cm	$2\pi r = 2 \times \frac{22}{7} \times 3.5 = 22$
7	**(d)** 21 cm	$r = \frac{C}{2\pi} = \frac{132 \times 7}{44} = 21$
8	**(a)** 44 cm	$\frac{180}{360} \times 2\pi r = 44$
9	**(a)** 6 cm^2	$s = 6$, area $= \sqrt{6 \cdot 3 \cdot 2 \cdot 1} = 6$
10	**(d)** 84 cm^2	$s = 24$, area $= \sqrt{24 \cdot 14 \cdot 7 \cdot 3} = 84$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$5x + 30 = 180 \Rightarrow x = 30$; $\angle A = 90^\circ$.	1+1
12	$2 \times 50^\circ = 100^\circ$.	2
13	Height $= \sqrt{25 - 9} = 4$ cm; area $= \frac{1}{2} \times 6 \times 4 = 12$ cm^2.	1+1
14	$s = 21$; area $= \sqrt{21 \times 14 \times 6 \times 1} = \sqrt{1764} = 42$ cm^2.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	At the point where the perpendicular bisectors of the segments joining the friends meet (the circumcentre) (2). Points on a perpendicular bisector are equidistant from the ends, so this point is equidistant from all three (1).	2+1
16	Area $= 144$ cm^2 (1). Side $= \sqrt{144} = 12$ cm (1). (Construction: the mean proportional of 16 and 9.)	2+1
17	The angle at the origin is 90°, so the hypotenuse joining $(6, 0)$ and $(0, 8)$ is a diameter (1). Its midpoint $(3, 4)$ is the centre (1); radius $= \frac{1}{2}\sqrt{36 + 64} = 5$ (1).	1+1+1
	OR: Let $AB = CD$. In $\triangle AOB$ and $\triangle COD$: $OA = OC$, $OB = OD$, $AB = CD$ ⇒ congruent (SSS) (2). So $\angle AOB = \angle COD$ (1).	2+1
18	Diagonal BD splits the parallelogram into two congruent triangles (1), so $\text{ar}(ABD) = 42$ cm^2 (2).	1+2
	OR: Arc $= \frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = 22$ cm (2). Perimeter $= 22 + 21 + 21 = 64$ cm (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Opposite angles of a parallelogram are equal and in a cyclic quadrilateral they add to 180°, so each is 90°: a rectangle (3). Each diagonal then subtends 90° at the other two vertices, so it is a diameter (2).	3+2
	OR: (i) 10 m rope: $\sqrt{169 - 25} = 12$ m; 24 m rope: $\sqrt{169 - 144} = 5$ m (2). (ii) $12 + 5 = 17$ m (1). (iii) $12 - 5 = 7$ m (1). (iv) 24 m rope is longer and nearer to the centre (1).	2+1+1+1
20	(i) $1 \times 19 = 19$, $2 \times 18 = 36$, $3 \times 17 = 51$, $4 \times 16 = 64$, $5 \times 15 = 75$, $6 \times 14 = 84$, $7 \times 13 = 91$, $8 \times 12 = 96$, $9 \times 11 = 99$, $10 \times 10 = 100$ (3). (ii) Greatest 100 (10×10), least 19 (1×19) (1). (iii) It is a square (1).	3+1+1
	OR: (i) If the diameters are $d_1, \ldots, d_n$ then $d_1 + \cdots + d_n = PQ$; length $= \pi(d_1 + \cdots + d_n)/2 = \pi PQ/2$, the same as the big semicircle (3). (ii) $\frac{22}{7} \times \frac{28}{2} = 44$ cm; also $\frac{22}{7} \times \frac{7+7+14}{2} = 44$ cm ✓ (2).	3+2

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ANSWER KEY & MARKING SCHEME SET D

Chapters 5 & 6: **Circles, Measuring Space: Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 105°	opposite angles of a cyclic quadrilateral add to 180°
2	**(c)** 57°	opposite angles of a cyclic quadrilateral add to 180°
3	**(b)** bisects the chord	perpendicular from the centre bisects the chord
4	**(b)** infinitely many	centre can be anywhere on the perpendicular bisector
5	**(d)** the centre of the circle	centre is equidistant from the chord's endpoints
6	**(d)** 180 cm^2	area $= bh$
7	**(a)** 12 sq. units	$s = 7$; $\sqrt{(s-a)(s-b)(s-c)(s-d)} = 12$
8	**(c)** 616 cm^2	$\pi r^2 = \frac{22}{7} \times 14^2 = 616$
9	**(b)** equal area	equal bases and the same height
10	**(a)** approximate (approaching)	he realised the ratio could not be exact

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\angle A + \angle C = 180^\circ$, so by the converse (Theorem 12) the vertices are concyclic.	1+1
12	$\angle BCD = 180^\circ - 70^\circ = 110^\circ$; opposite angle $\angle A = 180^\circ - 110^\circ = 70^\circ$.	1+1
13	The hexagon's perimeter is $6r = 3 \times (2r) = 3D$, and the circle is longer than the polygon inside it, so $C > 3D$, i.e. $\pi > 3$.	1+1
14	Arc $= \frac{1}{4} \times 2 \times \frac{22}{7} \times 14 = 22$ cm; perimeter $= 22 + 28 = 50$ cm.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Central angle is double: $6x - 10 = 2(2x + 5) = 4x + 10$ (2) $\Rightarrow x = 10$; angles 50° and 25° (1).	2+1
16	A triangle is a cyclic 4-gon with $d = 0$ (1). Then $s = \frac{a+b+c}{2}$ and $\sqrt{(s-a)(s-b)(s-c)(s-0)} = \sqrt{s(s-a)(s-b)(s-c)}$ (2), Heron's formula.	1+2
17	$4k + 5k = 180^\circ \Rightarrow k = 20^\circ$ (2); $\angle A = 80^\circ, \angle C = 100^\circ$ (1).	2+1
	OR: A line meets a circle in at most two points; hence three points on a line cannot all be on one circle (equivalently, a circle can be drawn through three points only if they are non-collinear).	1+2
18	30 min = half a turn: arc $= \pi r = 44$ cm (1). Area $= \frac{1}{2}\pi r^2 = \frac{1}{2} \times \frac{22}{7} \times 196 = 308$ cm^2 (2).	1+2
	OR: Half-perimeter 18; the maximum area occurs for the square: side 9 cm (2); area 81 cm^2 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Diagonal $= \sqrt{25 + 144} = 13$; angle in a semicircle ⇒ it is a diameter, radius 6.5 (2). (ii) $2 \times \frac{1}{2} \times 5 \times 12 = 60$ sq. units (2). (iii) Its sides in order are 5,5,12,12 (a kite) — opposite sides are not equal, so it is not a rectangle (1).	2+2+1
	OR: (i) The arc on the other side of the diameter subtends 180° at the centre, so the inscribed angle is 90° (2). (ii) Centre: $2 \times 62^\circ = 124^\circ$ (1). ACBD is cyclic so $\angle ADB = 180^\circ - 62^\circ = 118^\circ$ (2).	2+1+2
20	(i) Construct a square with the same area as the rectangle (1). (ii) Area $= 100$ cm^2; side $= 10$ cm (2). (iii) Square perimeter $= 40$ cm; rectangle perimeter $= 58$ cm — the square has the smaller perimeter for the same area (2).	1+2+2
	OR: (i) $\frac{22}{7} \times 1225 = 3850$ m^2 (1). (ii) $\frac{22}{7} \times 49 = 154$ m^2 (1). (iii) $3850 - 154 = 3696$ m^2 (1). (iv) $2\pi(35) + 2\pi(7) = 220 + 44 = 264$ m (2).	1+1+1+2

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